

Reciprocal-free divisors of prescribed order over finite fields and the saturation cost of cyclic quantum synchronizable codes

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Abstract

For a prime power q and an integer $N \geq 2$ with $\gcd(N, q) = 1$, let $w_0(N, q)$ be the classical least degree of a divisor of $x^N - 1$ of order N . We study its reciprocal-free restriction $w(N, q)$, the least degree of such a divisor h with $\gcd(h, h^*) = 1$, where h^* is the reciprocal polynomial, a constraint satisfied by saturating factors in cyclic quantum synchronizable codes. Always $w \geq w_0$; w may be larger or infinite. For odd N , reciprocal-free admissibility is determined by the 2 -adic valuations of the orders $\text{ord}_p(q)$ on the prime support, and a support-reduction theorem expresses $w(N, q)$ as a minimum over partitions of the prime support of N , where each block may use further prime divisors of N , to exponent 1 , as helpers; forbidding helpers recovers the classical formula for w_0 . For $N = p^a r^b$ (p, r distinct odd primes) we obtain an exact formula, and show that w/w_0 is unbounded over pairs (N, q) in which q varies with N . In the semisimple cyclic framework, every saturating factor has degree at least w , giving $K + 2d(D) \leq N + 2 - 2w$, with K the code dimension and $d(D)$ the classical minimum distance of the larger cyclic code, and an exact decomposition of the slack into a structural excess and a Singleton defect. A calibration of Class 1 of a published construction shows a construction-dependent excess of at least w in the synchronisation-factor degree. The results were stress-tested in 409,075 computational checks, with no failures. In the literature search performed for this work, no invariant equivalent to $w(N, q)$ was identified.

Keywords: Finite fields, Polynomial order, Reciprocal polynomials, Cyclic codes, Cyclotomic cosets, Quantum synchronizable codes

MSC Classification: 11T06 , 11T71 , 94B15 , 81P70 , 11A07

1 Introduction

1.1 Context

Let q be a prime power and $N \geq 2$ an integer. A polynomial $f \in F_q[x]$ with $f(0) \neq 0$ has *order* N if N is the least positive integer e with $f \mid x^e - 1$. Polynomials of prescribed order appear throughout the theory of finite fields: the least period of a linear recurring sequence equals the order of its minimal polynomial [1], the order of an invertible matrix over F_q equals the order of its minimal polynomial, and the order of a divisor of $x^N - 1$ controls the cyclic structure of the codes it generates.

The last point has a concrete role in the cyclic construction of quantum synchronizable codes (QSCs) of Fujiwara, Tonchev and Wong [2]. There, nested cyclic codes $C \subsetneq D$ of length N , with C dual-containing, yield a QSC whose misalignment tolerance is bounded by the order of the quotient $h = g_C/g_D$ of their generator polynomials; the tolerance is maximal exactly when $\text{ord}(h) = N$. This paper is primarily a study of divisors of $x^N - 1$ of prescribed order under a reciprocal-free constraint; the QSC setting supplies the motivation for that constraint and an application.

1.2 Classical baseline

The unrestricted problem asks for the least degree of a polynomial of order N :

$$w_0(N, q) = \min \{ \deg h : h \mid x^N - 1, \text{ord}(h) = N \}.$$

For $\gcd(N, q) = 1$, an irreducible factor of $x^N - 1$ of order $m \mid N$ has degree $\phi(m) = \text{ord}_m(q)$, and orders of coprime products combine by lcm, so $w_0(N, q)$ is the minimum of $\sum_{m \in S} \phi(m)$ over sets S of divisors of N with $\text{lcm } S = N$; it also has a partition form over the prime support of N (§3). The notation w_0 is ours, but the quantity is classical: it coincides with the least order of a linear recurrence of period N [1, 3], and with the least k for which $GL_k(q)$ contains an element of order N (§3.1).

1.3 The reciprocal-free restriction

Assume $\gcd(N, q) = 1$, and let $*$ denote the reciprocal polynomial. A cyclic code C of length N is dual-containing exactly when $g_C g_C^*$ divides $x^N - 1$. Since $x^N - 1$ is then squarefree, every divisor h of g_C satisfies $\gcd(h, h^*) = 1$. Hence the saturating quotient in the cyclic QSC construction is a reciprocal-free divisor of $x^N - 1$ of order N , which leads to

$$w(N, q) = \min \{ \deg h : h \mid x^N - 1, \text{ord}(h) = N, \gcd(h, h^*) = 1 \},$$

with $w(N, q) = \infty$ if no such h exists. The quantity w is not another name for w_0 : it minimises over a smaller class of divisors, so $w(N, q) \geq w_0(N, q)$. Equality can hold, the inequality can be strict, and w can be infinite; for example, $w(143, 3) = w_0(143, 3) = 8$, $w(55, 3) = 20$ whereas $w_0(55, 3) = 9$, and $w(225, 23) = 32$ whereas $w_0(225, 23) = 26$. In the literature search performed for this work, no invariant equivalent to $w(N, q)$ was identified.

1.4 Research question

For $\gcd(N, q) = 1$, how does the reciprocal-free constraint reshape the prescribed-order minimum-degree problem? What is the exact arithmetic structure of $w(N, q)$, and how does it differ from $w_0(N, q)$? For odd N we answer this structurally: $w(N, q)$ is reduced to an explicit finite optimisation over the prime support of N , and the constraint enters only through 2-adic invariants of the multiplicative orders of q .

1.5 Main results

The principal results are the following.

1. **Admissibility for odd moduli (Lemma 5.1).** For odd $p \nmid q$ let $t_p = v_2(\text{ord}_p(q))$. For odd $m \geq 3$ with $\gcd(m, q) = 1$, $-1 \notin \langle q \rangle \bmod m$ exactly when the classes t_p of the primes dividing m are not all equal to one and the same positive integer; hence admissibility depends only on the prime support of m . This criterion is known in substance; we give a complete proof.
2. **Support reduction (Theorem A; primary result).** For odd N with $\gcd(N, q) = 1$, $w(N, q)$ is the minimum, over partitions of the prime support of N , of a sum of block costs. Each block carries its full prime powers and may use further prime divisors of N to exponent 1 as helpers, solely to make its support admissible. Forbidding helpers recovers the classical partition formula for w_0 .
3. **Two prime powers (Theorem B).** For $N = p^a r^b$, with p and r distinct odd primes and $\gcd(N, q) = 1$, $w(N, q)$ has an exact four-case formula determined by t_p and t_r .
4. **Separation (Theorem C).** If $t_p \neq t_r$ and both are positive, then $w(\text{pr}, q) = \text{ord}_{pr}(q)$, while $w_0(\text{pr}, q) \leq \text{ord}_p(q) + \text{ord}_r(q)$. An explicit family of triples (p, r, q) with $p \geq 7$, in which the prime q is chosen depending on p and r , shows that w/w_0 is unbounded over pairs (N, q) .
5. **Saturation cost (Proposition 9.1, Corollary 9.2, Proposition 10.1).** In the semisimple cyclic QSC framework every saturating factor h satisfies $\deg h \geq w(N, q)$. Consequently $K + 2d(D) \leq N + 2 - 2w$, where $d(D)$ is the classical minimum distance of the larger code D , and the slack satisfies the exact identity $\Delta = 2(s - w) + 2(B + 1 - d(D))$, with $s = \deg h$ and $B = \deg g_D$.
6. **Calibration (Theorem 11.1).** For Class 1 of the construction of Li and Zhu [4] (q odd, $N = 2n$ with n an odd prime), $s \geq 2w$ and $\Delta \geq 2w$. Class 2 of [4] is examined only computationally, under a reconstructed index rule.

1.6 Scope and limitations

All results concerning w assume $\gcd(N, q) = 1$. Theorem A concerns odd N , Theorem B two distinct odd prime-power factors, and the unboundedness in Theorem C varies q together with N . We claim no result for fixed q , no asymptotic or density statement, no closed formula for three or more prime factors, no determination of the achievable QSC parameters, and no statement that the inequality involving w is sharp or stronger than Singleton-type bounds in general. Repeated-root constructions, including those of Du et al. [5], lie outside the setting in which w describes saturation, and are not

calibrated here. A finite computational record stress-tests the statements; it does not replace the proofs (§12).

1.7 Related literature

Quantum synchronizable codes were introduced by Fujiwara [6], and the cyclic framework used in this paper is that of [2]. Further constructions include codes from finite geometries [7], from quadratic residue codes and their supercodes [8], q -ary chain-containing codes [9], algebraic constructions [10], a class of codes from duadic codes with the highest possible tolerance against misalignment (T. Zhang and G. Ge, preprint arXiv:1508.00974, 2015), two families given in [11], and constructions from repeated-root cyclic codes [12] and repeated-root quasi-cyclic codes [13]. The constructions calibrated or excluded in this paper are those of [4, 5].

Dual-containing classical codes are the standard input to constructions of quantum codes from classical codes [14–16]. Negacyclic codes, which appear in the construction of [4], are studied for instance in [17, 18]. Reciprocal polynomials also govern cyclic codes with complementary duals [19, 20], as well as self-dual and self-orthogonal cyclic codes [21, 22]. The condition $-1 \in \langle q \rangle \bmod m$ is equivalent to m dividing $q^k + 1$ for some k ; divisors of numbers of the form $a^k + b^k$ are studied in [23].

Element orders in linear groups over finite fields have a separate literature, for example on element orders in groups related to the general linear group [24] and on the order of a typical matrix with entries in a finite field [25]. These works address questions different from the minimum-degree problem considered here.

Organisation. Section 2 collects preliminaries and the cyclic QSC framework; §3 treats the classical quantity w_0 ; §4 introduces reciprocal-free admissibility and w , with its lcm-cover form; §5 proves the admissibility criterion for odd moduli; §6 proves the support-reduction theorem; §7 gives the two-prime-power formula; §8 establishes the separation from w_0 ; §9 and §10 give the QSC application and the deficit decomposition; §11 calibrates the construction of [4]; §12 discusses scope and the computational verification; §13 concludes.

2 Preliminaries

Throughout, q is a prime power, F_q is the finite field with q elements, and $N \geq 1$ is an integer. For a positive integer m with $\gcd(m, q) = 1$, $\langle q \rangle$ denotes the cyclic subgroup of $(\mathbb{Z}/m\mathbb{Z})^\times$ generated by $q \bmod m$. All polynomials are in $F_q[x]$. “Divisor of $x^N - 1$ ” means a monic divisor.

2.1 Polynomial order and the lcm law

Definition (order) Let $f \in F_q[x]$ be nonconstant with $f(0) \neq 0$. The *order* $\text{ord}(f)$ is the least positive integer e such that $f(x)$ divides $x^e - 1$. For a nonzero constant c , set $\text{ord}(c) = 1$.

The order exists because x is a unit in the finite ring $F_q[x]/(f)$, since $f(0) \neq 0$. In fact $\text{ord}(f)$ is the multiplicative order of the class of x in that ring. Consequently, for every positive integer e ,

$$f(x) \mid x^e - 1 \iff \text{ord}(f) \mid e. \quad (2.1)$$

For irreducible f , $\text{ord}(f)$ equals the multiplicative order of any root of f in a splitting field [1].

Lemma 2.1 (lcm law) *Let $f, g \in F_q[x]$ satisfy $f(0)g(0) \neq 0$ and $\gcd(f, g) = 1$. Then*

$$\text{ord}(fg) = \text{lcm}(\text{ord}(f), \text{ord}(g)).$$

Proof Since f and g are coprime, fg divides $x^e - 1$ if and only if both f and g divide $x^e - 1$. By (2.1) this holds if and only if $\text{ord}(f) \mid e$ and $\text{ord}(g) \mid e$, that is, $\text{lcm}(\text{ord}(f), \text{ord}(g)) \mid e$. The least such e is $\text{lcm}(\text{ord}(f), \text{ord}(g))$. $\square$

By induction, the order of a product of pairwise coprime polynomials with nonzero constant terms is the lcm of their orders.

2.2 Cyclotomic cosets and reciprocal polynomials

Assume $\gcd(m, q) = 1$. The q -cyclotomic coset of $i \in \mathbb{Z}/m\mathbb{Z}$ is $C_i^{(m)} = \{ i q^j \bmod m : j \geq 0 \}$; the cosets partition $\mathbb{Z}/m\mathbb{Z}$. For $m \geq 1$ write $\text{o}(m) = \text{ord}_m(q)$ for the multiplicative order of q modulo m , with $\text{o}(1) = 1$.

Factorisation of $x^N - 1$. Let $\gcd(N, q) = 1$ and let ξ be a primitive N -th root of unity in an extension of F_q . Then $x^N - 1$ is squarefree, and

$$x^N - 1 = \prod_C M_C(x), \text{ where } M_C(x) = \prod_{j \in C} (x - \xi^j),$$

the product running over the q -cyclotomic cosets C modulo N ; each M_C is irreducible over F_q , and every irreducible factor of $x^N - 1$ arises this way exactly once [1]. If $C = C_i^{(N)}$, the roots of M_C have multiplicative order $m = N/\gcd(i, N)$, so $\text{ord}(M_C) = m \mid N$; and an element α of order m satisfies $\alpha^{q^k} = \alpha$ if and only if $m \mid q^k - 1$, so $\deg M_C = |C| = \text{o}(m)$. Thus **every irreducible factor of $x^N - 1$ of order m has degree $\text{o}(m)$** , and for every $m \mid N$ such a factor exists, for instance the minimal polynomial of $\xi^{N/m}$.

For $f \in F_q[x]$ with $f(0) \neq 0$, the (monic) reciprocal is $f(x) = f(0)^{-1} x^{\deg f} f(1/x)$. Its roots are the inverses of the roots of f , with multiplicities; $f \mapsto f^*$ preserves degree and irreducibility, $(fg)^* = fg$, and $\text{ord}(f) = \text{ord}(f^*)$. We call f self-reciprocal* if $f^* = f$. Since $(x^N - 1)^* = x^N - 1$, the reciprocal of a divisor of $x^N - 1$ is again such a divisor.

Lemma 2.2 (self-reciprocal irreducible factors) *Let $\gcd(m, q) = 1$, and let $f \in F_q[x]$ be irreducible with $\text{ord}(f) = m$. Then*

$$f \text{ is self-reciprocal} \iff -1 \in \langle q \rangle \leq (\mathbb{Z}/m\mathbb{Z})^\times.$$

Proof Let α be a root of f . Then α has multiplicative order m , and the roots of f are α^{q^j} , $j \geq 0$. Since f and f^* are monic irreducible, $f^* = f$ holds if and only if the root α^{-1} of f^* is a root of f . That is, $\alpha^{-1} = \alpha^{q^j}$ for some j , which is equivalent to $q^j \equiv -1 \pmod{m}$. $\square$

For $m \in \{1, 2\}$ one has $-1 \equiv 1 \pmod{m}$. Lemma 2.2 therefore recovers the fact that $x - 1$ and $x + 1$ (orders 1 and 2) are self-reciprocal. The criterion is classical in substance; we include the proof for completeness.

2.3 Cyclic codes, duals, and dual containment

Let $R_N = F_q[x]/(x^N - 1)$. A vector $(c_0, \dots, c_{N-1}) \in F_q^N$ is identified with $c(x) = \sum c_i x^i \in R_N$.

A *cyclic code* of length N is an ideal C of R_N . It is generated by a unique monic divisor g_C of $x^N - 1$, its *generator polynomial*, and $\dim C = N - \deg g_C$.

The *check polynomial* is

$$h_C = (x^N - 1)/g_C.$$

Two facts are used below: For monic divisors a, b of $x^N - 1$, $\langle a \rangle \subseteq \langle b \rangle$ in R_N if and only if $b \mid a$. $C^\perp$ denotes the dual of C with respect to the standard inner product.

Lemma 2.3 (dual containment) *Let $\gcd(N, q) = 1$, and let C be a cyclic code of length N with generator polynomial g_C . Then*

$$C^\perp \subseteq C \iff g_C(x) g_C^*(x) \mid x^N - 1.$$

The criterion is standard; for completeness, a proof (via the identification $C^\perp = \langle h_C^ \rangle$, where $h_C = (x^N - 1)/g_C$) is given in Online Resource 1.*

The semisimple hypothesis matters for the consequence we need. If $\gcd(N, q) = 1$, then $x^N - 1$ is squarefree. So $g_C g_C^ \mid x^N - 1$ forces $\gcd(g_C, g_C^*) = 1$. Every divisor h of g_C then also satisfies $\gcd(h, h) = 1$, because $h h$ divides $g_C g_C^*$. Without the semisimple hypothesis this implication can fail.*

Lemma 2.4 (Singleton bound) *Every linear $[N, k, d]$ code over F_q with $k \geq 1$ satisfies $d \leq N - k + 1$.*

Proof The codewords vanishing in the first $k - 1$ coordinates form a subspace of dimension at least 1. A nonzero element of that subspace has weight at most $N - (k - 1)$. $\square$

2.4 The cyclic QSC framework

We recall only the part of the cyclic QSC framework of Fujiwara, Tonchev and Wong [2] that is used later. Let $C \subsetneq D$ be cyclic codes of length N over F_q with generator polynomials g_C and g_D , where C is dual-containing ($C^\perp \subseteq C$) and $g_D \mid g_C$, $g_D \neq g_C$.

Put $h = g_C/g_D$ and $k_C = \dim C = N - \deg g_C$; then $h \mid x^N - 1$, so $\text{ord}(h) \leq N$ by (2.1). For all non-negative integers a_l, a_r with

$$a_l + a_r < \text{ord}(h), \quad (2.2)$$

[2] constructs an $(a_l, a_r) - QSC$ of length $N + a_l + a_r$ and dimension

$$K = 2k_C - N = N - 2 \deg g_C, \quad (2.3)$$

which corrects misalignments of up to a_l positions to the left and a_r to the right; its quantum error-correcting capabilities are expressed in [2] through the minimum distances of C and D . Throughout, $d(D)$ denotes the **classical** minimum distance of D , never a quantum distance. We call the pair (C, D) , or h , *saturated* if $\text{ord}(h) = N$; then (2.2) permits all $a_l + a_r \leq N - 1$. Only (2.2) and (2.3) are used later, and we make no claim that the quantities introduced below determine K , $d(D)$ or the set of achievable parameters. The constructions of [4, 5] are instances of this framework, discussed only in §11 and §12.

2.5 Notation

Throughout, the semisimple condition $\gcd(N, q) = 1$ is assumed wherever $w(N, q)$ or $w_0(N, q)$ appears; $\text{o}(m) = \text{ord}_m(q)$ for $\gcd(m, q) = 1$; $t_p = v_2(\text{o}(p))$ for an odd prime $p \nmid q$, where v_2 is the 2-adic valuation (used from §5 on). An integer $m \geq 1$ with $\gcd(m, q) = 1$ is *admissible* if $-1 \notin \langle q \rangle \leq (\mathbb{Z}/m\mathbb{Z})^\times$; every admissible m satisfies $m \geq 3$, and by Lemma 2.2 m is admissible if and only if the irreducible factors of order m are not self-reciprocal. A polynomial h with $h(0) \neq 0$ is *reciprocal-free* if $\gcd(h, h^*) = 1$. Minima over empty sets are $+\infty$.

3 Classical minimum degree w_0

Throughout this section, $\gcd(N, q) = 1$ and $N \geq 2$. Notation is as in §2.

3.1 Definition and basic interpretation

The *classical minimal degree of order N* is

$$w_0(N, q) = \min \{ \deg h : h \mid x^N - 1, \text{ord}(h) = N \}.$$

The divisibility condition imposes no restriction beyond the order condition: by (2.1), every polynomial of order N divides $x^N - 1$. So $w_0(N, q)$ is the least degree of any polynomial over F_q with order N . It is finite, because an irreducible factor of $x^N - 1$ of order N exists (§2.2).

The quantity has two classical interpretations.

Linear recurrences. The least period of a periodic linear recurring sequence equals the order of its minimal polynomial [1]. The set of least periods realised by linear recurrences of order k coincides with the set of orders of polynomials of degree

at most k [3]. Hence $w_0(N, q)$ is the least k for which some linear recurrence of order k over F_q has a sequence of least period N .

Matrices. $w_0(N, q)$ is the least k for which $GL_k(q)$ contains an element of order N .

Proof of the matrix interpretation Let $g \in GL_k(q)$ have minimal polynomial μ . Then $\mu(0) \neq 0$, and $g^e = I$ if and only if $\mu \mid x^e - 1$, so the order of g equals $\text{ord}(\mu)$. If g has order N , then μ has order N and degree at most k , so $w_0(N, q) \leq k$. Conversely, let h be a polynomial of order N with $\deg h = w_0(N, q)$. Its companion matrix lies in $GL_{\deg h}(q)$, has minimal polynomial h , and so has order N . □

3.2 lcm-cover and partition forms

Proposition 3.1 (classical forms of w_0) Let $\gcd(N, q) = 1$ and $N \geq 2$.

- (a) **lcm-cover form:** $w_0(N, q) = \min \{ \sum_{m \in S} o(m) : S \subseteq \{ d : d \mid N, d > 1 \}, \text{lcm } S = N \}$.
- (b) **Partition form:** if $N = \prod_{p \in P} p^{a_p}$, with P the set of prime divisors of N , and $n_A = \prod_{p \in A} p^{a_p}$ for nonempty $A \subseteq P$, then $w_0(N, q) = \min_{\mathcal{P}} \sum_{A \in \mathcal{P}} o(n_A)$, where $\mathcal{P}$ ranges over the partitions of P .

Part (a) is the unconstrained case of Proposition 4.2, and part (b) is formula (6.3); both are proved in §4.3 and §6.5 alongside their reciprocal-free counterparts. Proposition 3.1 is a direct consequence of classical facts, and is not claimed as new. Formula (b) is the unrestricted counterpart of Theorem A: §6 shows that the reciprocal-free restriction changes exactly the block costs, and nothing else.

The reciprocal-free restriction $w(N, q)$ of w_0 is formally introduced in Definition 4.1. Since it minimises over a subclass of the divisors allowed for w_0 , $w(N, q) \geq w_0(N, q)$. The inequality can be strict, or w can be infinite (Proposition 4.3(1) and §7.4).

4 Reciprocal-free admissibility and $w(N, q)$

Throughout this section, $\gcd(N, q) = 1$ and $N \geq 2$. So $x^N - 1$ is squarefree and factors as in §2.2.

4.1 Admissible divisors

A monic divisor h of $x^N - 1$ is called *admissible*, or *reciprocal-free*, if

$$\gcd(h, h^*) = 1.$$

Since $x^N - 1$ is squarefree, h is a product of distinct irreducible factors of $x^N - 1$, and h^* is the product of their reciprocals. So h is reciprocal-free exactly when two conditions hold: **(a)** no irreducible factor of h is self-reciprocal; **(b)** h does not contain both an irreducible factor f and its reciprocal $f^* \neq f$.

By Lemma 2.2, condition (a) says that every irreducible factor of h has an admissible order.

This is the condition met by every divisor of the generator polynomial of a dual-containing cyclic code (Remark after Lemma 2.3). Its existence aspect is classical in the theory of self-orthogonal and self-dual cyclic codes [21, 22].

4.2 Definition of $w(N, q)$

Definition 4.1 For $\gcd(N, q) = 1$ and $N \geq 2$ define

$$w_0(N, q) = \min \{ \deg h : h \mid x^N - 1, \text{ord}(h) = N \},$$

$$w(N, q) = \min \{ \deg h : h \mid x^N - 1, \text{ord}(h) = N, \gcd(h, h^*) = 1 \},$$

with $w(N, q) = \infty$ if no such h exists.

$w_0(N, q)$ is the classical unconstrained quantity: the least degree of a divisor of $x^N - 1$ of order N . It is always finite, since an irreducible factor of order N exists. $w(N, q)$ is its restriction to reciprocal-free divisors. The only difference between the two is the constraint $\gcd(h, h^*) = 1$.

4.3 The lcm-cover form

Proposition 4.2 (lcm-cover form) Let $\gcd(N, q) = 1$ and $N \geq 2$. Then

$$w(N, q) = \min_S \sum_{m \in S} o(m),$$

where S ranges over the sets of admissible divisors m of N with $\text{lcm}_{m \in S} m = N$. Likewise,

$$w_0(N, q) = \min_S \sum_{m \in S} o(m),$$

where S ranges over the sets of divisors $m > 1$ of N with $\text{lcm}_{m \in S} m = N$.

Proof We prove the statement for w . The proof for w_0 is identical with every admissibility condition removed.

($\leq$) *Every admissible cover gives a feasible h .* Let S be a set of admissible divisors of N with $\text{lcm } S = N$. For each $m \in S$ choose an irreducible factor f_m of $x^N - 1$ with $\text{ord}(f_m) = m$ (§2.2). Distinct m give distinct f_m , so $h = \prod_{m \in S} f_m$ divides the squarefree polynomial $x^N - 1$. By Lemma 2.1, $\text{ord}(h) = \text{lcm } S = N$. h is reciprocal-free: Each f_m^* is irreducible of order m . A common irreducible factor of h and h^* would be $f_m = f_{m'}^*$ for some $m, m' \in S$. Comparing orders gives $m = m'$, so f_m would be self-reciprocal. That contradicts Lemma 2.2, since m is admissible. By §2.2, $\deg h = \sum_{m \in S} \deg f_m = \sum_{m \in S} o(m)$. Hence $w(N, q) \leq \sum_{m \in S} o(m)$.

($\geq$) *Every feasible h gives an admissible cover.* Let h be feasible for $w(N, q)$. Write $h = f_1 \cdots f_r$ as a product of distinct irreducible factors of $x^N - 1$, with $m_i = \text{ord}(f_i)$ dividing N . If some f_i were self-reciprocal, it would divide $\gcd(h, h^*)$. So by Lemma 2.2 every m_i is admissible. By Lemma 2.1, $\text{lcm}(m_1, \dots, m_r) = \text{ord}(h) = N$. Let $S = \{m_1, \dots, m_r\}$, viewed as a set. Then $\text{lcm } S = N$, and since $\deg f_i = o(m_i)$,

$$\sum_{m \in S} o(m) \leq \sum_{i=1}^r \deg f_i = \deg h.$$

Taking h of minimal degree gives the reverse inequality. □

4.4 Elementary properties

Proposition 4.3 *Let $\gcd(N, q) = 1$ and $N \geq 2$.*

1. $w(N, q) < \infty$ if and only if N is admissible.
2. $w(N, q) \geq w_0(N, q)$.
3. If $w(N, q) < \infty$, then $1 \leq w(N, q) \leq \text{ord}_N(q)$.
4. If $N > 2$, then $w(N, q) = 1$ if and only if $N \mid q - 1$.

Proof (1) If N is admissible, $S = \{N\}$ is admissible with $\text{lcm } N$, so $w(N, q) \leq o(N) < \infty$ by Proposition 4.2.

Conversely, suppose N is not admissible, so $q^j \equiv -1 \pmod{N}$ for some j . Reducing modulo any divisor m of N gives $q^j \equiv -1 \pmod{m}$, so no divisor of N is admissible. Every S with $\text{lcm } S = N \geq 2$ is nonempty, so no admissible S exists. Hence $w(N, q) = \infty$.

(2) Every h feasible for $w(N, q)$ is feasible for $w_0(N, q)$.

(3) A feasible h has $\text{ord}(h) = N \geq 2$, so h is nonconstant and $\deg h \geq 1$. The upper bound is the choice $S = \{N\}$ in (1).

(4) Suppose $N \mid q - 1$. Then $o(N) = 1$ and $\langle q \rangle = \{1\}$ in $(\mathbb{Z}/N\mathbb{Z})^\times$. Since $N > 2$, $-1 \not\equiv 1 \pmod{N}$, so N is admissible. By (1) and (3), $w(N, q) = 1$.

Conversely, suppose $w(N, q) = 1$. Then some feasible $h = x - a$ has degree 1. So $a \in F_q^\times$ has multiplicative order $\text{ord}(h) = N$, which forces $N \mid q - 1$. □

The restriction $N > 2$ in (4) is necessary: the only divisors of $x^2 - 1$ of order 2 are $x + 1$ and $x^2 - 1$, neither reciprocal-free, so $w(2, q) = \infty$, in accordance with (1).

4.5 Comparison with w_0

Both quantities are minimum-cost lcm-covers of N by divisors with cost $o(m)$; they differ only in that w allows admissible divisors alone, so $w \geq w_0$. Equality can occur: if $N > 2$ and $N \mid q - 1$, then $w(N, q) = w_0(N, q) = 1$. Equality need not occur, even when w is finite. For $q = 3$ and $N = 5$, $3^2 \equiv -1 \pmod{5}$, so 5 is not admissible and $w(5, 3) = \infty$, while $w_0(5, 3) = o(5) = 4$. For $q = 3$ and $N = 55$, 5 is not admissible, whereas 11 is admissible since $\langle 3 \rangle = \{1, 3, 4, 5, 9\} \pmod{11}$ does not contain -1 ; hence 55 is admissible (reduction mod 11). Any cover of 55 must contain a divisor divisible by 5, and the only admissible one is 55, so $w(55, 3) = o(55) = \text{lcm}(4, 5) = 20$, while $S = \{5, 11\}$ gives $w_0(55, 3) = o(5) + o(11) = 4 + 5 = 9$. The quantitative separation between w and w_0 is studied in §8.

5 Admissibility structure for odd N

Throughout this section q is a prime power. No parity assumption is made on q .

5.1 2-adic classes t_p

For an odd prime $p \nmid q$, define the *2-adic class* of p (relative to q) as

$$t_p = v_2(o(p)) = v_2(ord_p(q)),$$

where v_2 is the 2-adic valuation. This is a standard arithmetic quantity; the notation is fixed only for later use.

The class of p also controls every power of p . For $a \geq 1$ we have

$$v_2(o(p^a)) = t_p, \quad (5.1)$$

proved in Step 2 of §5.2. So any criterion expressed through the 2-adic valuations of the orders $o(p^{a_p})$ depends only on the primes p , not on the exponents a_p . This is the source of the support invariance in §5.3.

5.2 The 2-adic admissibility criterion

Lemma 5.1 (2-adic admissibility criterion) *Let $m \geq 3$ be odd with $\gcd(m, q) = 1$. Then m is admissible, that is $-1 \notin \langle q \rangle \leq (\mathbb{Z}/m\mathbb{Z})^\times$, if and only if it is **not** the case that all t_p (p a prime divisor of m) equal one and the same positive integer.*

Equivalently, m is inadmissible if and only if $t_p \geq 1$ for every prime $p \mid m$ and $t_p = t_r$ for every pair of prime divisors p, r of m .

The condition $t_p \geq 1$ cannot be dropped from the pairwise form: when m is a prime power, the pairwise condition is vacuous.

Proof Write $m = \prod_{p \mid m} p^{a_p}$ with $a_p \geq 1$, and put $o_p = o(p^{a_p})$.

Step 1 (CRT reduction). By the Chinese remainder theorem, $q^j \equiv -1 \pmod{m}$ holds if and only if $q^j \equiv -1 \pmod{p^{a_p}}$ for every $p \mid m$. Hence

$$-1 \in \langle q \rangle \pmod{m} \iff \text{there is a single exponent } j \geq 0 \text{ with } q^j \equiv -1 \pmod{p^{a_p}} \text{ for every } p \mid m. \quad (5.2)$$

The reduction is to the prime powers p^{a_p} , not to the primes p . The passage to primes happens only through (5.1).

Step 2 (proof of (5.1)). Reducing modulo p shows $o(p) \mid o(p^a)$. Write $q^{o(p)} = 1 + p u$ with $u \in \mathbb{Z}$. The binomial theorem gives $(1 + p u)^{p^{a-1}} \equiv 1 \pmod{p^a}$, so $o(p^a) \mid o(p) p^{a-1}$. Hence $o(p^a)/o(p)$ divides p^{a-1} , which is odd. So $v_2(o(p^a)) = v_2(o(p)) = t_p$.

Step 3 (a single prime power). Since p is odd, $(\mathbb{Z}/p^a\mathbb{Z})^\times$ is cyclic, and it contains a unique element of order 2, namely -1 (note $-1 \not\equiv 1$ because $p^a \geq 3$). The subgroup $\langle q \rangle \pmod{p^a}$ is cyclic of order o_p . It contains an element of order 2 if and only if o_p is even, and that element is then $q^{o_p/2} = -1$. Consequently $q^j \equiv -1 \pmod{p^{a_p}}$ if and only if o_p is even and

$$j \equiv o_p/2 \pmod{o_p}. \quad (5.3)$$

By (5.1), o_p is even if and only if $t_p \geq 1$. Writing $o_p = 2^{t_p} u_p$ with u_p odd, (5.3) reads $j \equiv 2^{t_p-1} u_p \pmod{2^{t_p} u_p}$.

Step 4 (necessity). Suppose $-1 \in \langle q \rangle \pmod m$, and let j be as in (5.2). By Step 3, every $t_p \geq 1$, and $j = 2^{t_p-1} u_p + k_p 2^{t_p} u_p = 2^{t_p-1} u_p (1 + 2k_p)$ for some integer k_p . Since $u_p (1 + 2k_p)$ is odd, $v_2(j) = t_p - 1$. Therefore $t_p = v_2(j) + 1$ for every $p \mid m$. So all t_p equal the same positive integer.

Step 5 (sufficiency: a common exponent). Suppose $t_p = t \geq 1$ for every $p \mid m$. Then $o_p = 2^t u_p$ with u_p odd. Let $U = \text{lcm}_{p \mid m} u_p$, which is odd, and put

$$j = 2^{t-1} U.$$

For each p , $j = (o_p/2) \cdot (U/u_p)$, and U/u_p is an odd integer, say $2k + 1$. Hence $j = o_p/2 + k o_p \equiv o_p/2 \pmod{o_p}$. By Step 3, $q^j \equiv -1 \pmod{p^{a_p}}$ for every $p \mid m$. By (5.2), $-1 \in \langle q \rangle \pmod m$.

Steps 4 and 5 prove the lemma. The criterion involves only the values t_p , and by (5.1) these depend only on the prime divisors of m , not on their exponents. $\square$

5.3 Support invariance

For an integer $m \geq 1$ let $\text{supp}(m)$ denote the set of prime divisors of m .

Corollary 5.2 (support invariance) *Let m and m' be odd integers ≥ 3 that are coprime to q , with $\text{supp}(m) = \text{supp}(m')$. Then m is admissible if and only if m' is admissible.*

Proof By Lemma 5.1, the admissibility of m depends only on the family $(t_p)_{p \in \text{supp}(m)}$, and the same holds for m' . $\square$

Consequence used in §6. Let N be odd with $\gcd(N, q) = 1$, and let $m' \mid m \mid N$ with $\text{supp}(m') = \text{supp}(m)$. Then: m' is admissible if and only if m is admissible; $o(m')$ divides $o(m)$, because $q^{o(m)} \equiv 1 \pmod m$ implies $q^{o(m)} \equiv 1 \pmod{m'}$.

So a divisor can be shrunk, provided its prime support is kept, without losing admissibility and without increasing its cost $o(\cdot)$.

Remark on even m . Lemma 5.1 and Corollary 5.2 are stated and proved only for odd m . Step 3 uses the cyclicity of $(\mathbb{Z}/p^a \mathbb{Z})^\times$ for odd p , and this fails for $(\mathbb{Z}/2^a \mathbb{Z})^\times$ when $a \geq 3$. Accordingly, the structural results of §6 are restricted to odd N . Even moduli are not treated in this paper.

6 Support-reduction theorem

Throughout this section: $N \geq 3$ is odd with $\gcd(N, q) = 1$; $N = \prod_{p \in P} p^{a_p}$, with $P = \text{supp}(N) \neq \emptyset$ and $a_p \geq 1$; by a *partition* of P we mean a set $\{A_1, \dots, A_s\}$ of pairwise disjoint nonempty subsets of P whose union is P .

6.1 Blocks and helper primes

For a nonempty *block* $A \subseteq P$ put

$$n_A = \prod_{p \in A} p^{a_p}.$$

Helper sets. A *helper set* for A is a subset $H \subseteq P \setminus A$. Each helper prime is thus a prime divisor of N that lies outside A , and it is used to exponent 1 only. Set

$$m(A, H) = n_A \prod_{r \in H} r.$$

Since every $r \in H$ lies in P and $1 \leq a_r$, we have

$$m(A, H) \mid N \text{ and } \text{supp}(m(A, H)) = A \cup H. \quad (6.1)$$

The helper set H is *admissible* for A if $m(A, H)$ is admissible. By Corollary 5.2 this depends only on $A \cup H$.

Block cost. Define

$$c(A) = \min \{ o(m(A, H)) : H \subseteq P \setminus A, m(A, H) \text{ admissible} \},$$

with $c(A) = \infty$ if no admissible helper set exists. The minimum is over the finite family of subsets of $P \setminus A$, so it is attained whenever that family contains an admissible helper set.

In particular, $m(A, H)$ is always a divisor of N , hence the order of an irreducible factor of $x^N - 1$ (§2.2).

6.2 Statement of Theorem A

Theorem A (support-reduction formula) Let $N = \prod_{p \in P} p^{a_p}$ be odd with $\gcd(N, q) = 1$. With n_A , $m(A, H)$ and $c(A)$ as in §6.1,

$$w(N, q) = \min_{\{A_1, \dots, A_s\} \text{ partition of } P} \sum_{j=1}^s c(A_j), \quad (6.2)$$

where both sides may equal ∞ . Moreover, with the block cost $c_0(A) = o(n_A)$ (no helpers),

$$w_0(N, q) = \min_{\{A_1, \dots, A_s\} \text{ partition of } P} \sum_{j=1}^s o(n_{A_j}). \quad (6.3)$$

Denote the right-hand side of (6.2) by $F(N, q)$. We prove $F \leq w$ in §6.3 and $w \leq F$ in §6.4, and then (6.3) in §6.5.

6.3 Proof: reduction of an optimal cover ($F \leq w$)

If $w(N, q) = \infty$ there is nothing to prove. Assume $w(N, q) < \infty$.

Starting point. By Proposition 4.2, fix a set S of admissible divisors of N with $\text{lcm } S = N$ and $\sum_{m \in S} o(m) = w(N, q)$. Such an S arises from an optimal reciprocal-free divisor h by collecting the orders of its irreducible factors (proof of Proposition 4.2).

Assigning each prime to a carrier. For each $p \in P$,

$$a_p = v_p(N) = \max_{m \in S} v_p(m),$$

because $\text{lcm } S = N$. Choose one element $\mu(p) \in S$ with $v_p(\mu(p)) = a_p$. For $m \in S$ set $A_m = \{ p \in P : \mu(p) = m \}$. Then: the nonempty sets among the A_m form a partition $\mathcal{P}$ of P ; each prime lies in exactly one block, because μ is a function.

No assumption is made that S has only one element per block, or one block per element. An element $m \in S$ may carry several primes (A_m may have several elements), or none ($A_m = \emptyset$).

Shrinking a carrier. Let $m \in S$ with $A_m \neq \emptyset$. Put $H_m = \text{supp}(m) \setminus A_m \subseteq P \setminus A_m$ and

$$m' = m(A_m, H_m) = n_{A_m} \prod_{r \in H_m} r.$$

Then: $m' \mid m$. For $p \in A_m$, $v_p(m') = a_p = v_p(m)$. For $r \in H_m$, $v_r(m') = 1 \leq v_r(m)$. No other prime divides m' . **Admissibility is preserved.** $\text{supp}(m') = A_m \cup H_m = \text{supp}(m)$. Since m is admissible, Corollary 5.2 shows m' is admissible. So H_m is an admissible helper set for A_m . **The cost does not increase.** By §5.3, $o(m') \mid o(m)$. Hence

$$c(A_m) \leq o(m') \leq o(m). \quad (6.4)$$

Why the construction is sound. *Prime powers are never split.* Each $p \in P$ is carried at its full power p^{a_p} by exactly one block, the block of $\mu(p)$. The other occurrences of p in S are replaced by the single power p , as helpers. This can only lower the cost, by (6.4). *The target N is still reached.* Every full power p^{a_p} divides $n_{A_{\mu(p)}}$. The helper powers r^1 satisfy $r^1 \mid r^{a_r}$, so they never exceed the exponent required in N . *Helpers serve admissibility only.* A helper r of a block A has its full power r^{a_r} supplied by a different block.

The bound. Discard the elements with $A_m = \emptyset$; this only decreases the sum. By (6.4),

$$F(N, q) \leq \sum_{A \in \mathcal{P}} c(A) = \sum_{m \in S, A_m \neq \emptyset} c(A_m) \leq \sum_{m \in S} o(m) = w(N, q).$$

6.4 Proof: realisation, including coinciding blocks ($w \leq F$)

If $F(N, q) = \infty$ there is nothing to prove. Otherwise fix a partition $\{A_1, \dots, A_s\}$ of P attaining $F(N, q)$. For each j fix an admissible helper set $H_j \subseteq P \setminus A_j$ with $o(m_j) = c(A_j)$, where $m_j = m(A_j, H_j)$.

The m_j form an admissible cover of N . (i) $m_j \mid N$ by (6.1). So each m_j is the order of an irreducible factor of $x^N - 1$. (ii) Each m_j is admissible. (iii) $\text{lcm}_j m_j = N$. For $p \in P$ there is exactly one j with $p \in A_j$, and then $v_p(m_j) = a_p$. For every i , $v_p(m_i) \leq a_p$ by (i).

Coinciding blocks. The integers $m_1, \dots, m_s$ need not be pairwise distinct a priori. For instance, when $a_p = a_r = 1$, the block $\{p\}$ with helper set $\{r\}$ and the block $\{r\}$ with helper set $\{p\}$ both give $m = pr$.

Let $S = \{m_1, \dots, m_s\}$ be the underlying set. By (i)–(iii), S is a set of admissible divisors of N with $\text{lcm } S = N$, and

$$\sum_{m \in S} o(m) \leq \sum_{j=1}^s o(m_j) = F(N, q), \quad (6.5)$$

with strict inequality if and only if two of the m_j coincide.

Constructing the divisor h . As in the proof of Proposition 4.2, choose for each $m \in S$ **one** irreducible factor f_m of $x^N - 1$ with $\text{ord}(f_m) = m$, and put $h = \prod_{m \in S} f_m$. Then:

1. $h \mid x^N - 1$. Each f_m is an irreducible factor of $x^N - 1$ of order $m \mid N$. Distinct m give distinct f_m , and $x^N - 1$ is squarefree.
2. **Only divisors of N occur as orders.** Every order used is some m_j , and $m_j \mid N$. Helper primes enter only as prime divisors of N at exponent 1.
3. $\text{ord}(h) = N$, by Lemma 2.1 and (iii).
4. h is **reciprocal-free**. A common irreducible factor of h and h^* would be $f_m = f_{m'}^*$ with $m, m' \in S$. Equal orders force $m = m'$, so f_m would be self-reciprocal, contradicting Lemma 2.2 because m is admissible.
5. $\deg h = \sum_{m \in S} o(m) \leq F(N, q)$, by §2.2 and (6.5).

Hence $w(N, q) \leq F(N, q)$. Combined with §6.3, this proves (6.2), including the case where both sides are infinite.

Clarification on coinciding blocks. No second irreducible factor of the same order is needed. A single factor f_m serves every block producing m . At an optimal partition with optimal helper sets, coincidences do not in fact occur. If $m_i = m_j$ for some $i \neq j$, then (6.5) would be strict, and the constructed h would give $w(N, q) < F(N, q)$. That contradicts §6.3. Consequently, for such a choice the m_j are pairwise distinct, and $\deg h = \sum_j c(A_j) = F(N, q)$ exactly.

6.5 Specialisation to w_0

Without the reciprocal-free constraint, Proposition 4.2 expresses $w_0(N, q)$ as the minimum of $\sum_{m \in S} o(m)$ over sets S of divisors $m > 1$ of N with $\text{lcm } S = N$. Admissibility plays no role, so helper primes are never needed.

Lower bound. Repeat §6.3 with $m' = n_{A_m}$ in place of $m(A_m, H_m)$. Then $m' \mid m$, so $o(n_{A_m}) \leq o(m)$. This gives $\min \sum o(n_{A_j}) \leq w_0(N, q)$.

Upper bound. For a partition $\{A_j\}$, the divisors n_{A_j} are pairwise distinct, since their supports are disjoint and nonempty. They satisfy $\text{lcm}_j n_{A_j} = N$. The construction of §6.4, without the reciprocal-free verification, gives a divisor of $x^N - 1$ of order N and degree $\sum_j o(n_{A_j})$.

This proves (6.3), which is a direct consequence of classical facts (§3.2). Equivalently, (6.3) is (6.2) with every helper set forced to be empty. In the unconstrained problem the empty helper set is always optimal, because $o(n_A) \mid o(m(A, H))$.

The difference between (6.2) and (6.3) lies entirely in the helper sets. They are required exactly for the blocks A whose own support is inadmissible, and by Lemma 5.1 this is decided by the 2-adic classes t_p .

6.6 Worked example: $N = 225$

Take $N = 225 = 3^2 \cdot 5^2$ and $q = 23$, so $P = \{3, 5\}$, with $o(3) = 2$, $o(5) = 4$, $o(9) = 6$, $o(25) = 20$, and, since $o(ab) = \text{lcm}(o(a), o(b))$ for coprime a, b , $o(45) = 12$, $o(75) = 20$, $o(225) = 60$. Here $t_3 = 1$ and $t_5 = 2$, so by Lemma 5.1 the supports $\{3\}$ and $\{5\}$ are inadmissible and $\{3, 5\}$ is admissible. Hence $c(\{3, 5\}) = o(225) = 60$; $c(\{3\})$ requires the helper 5, giving $m(\{3\}, \{5\}) = 45$ and $c(\{3\}) = 12$; and $c(\{5\})$ requires the helper 3, giving $m(\{5\}, \{3\}) = 75$ and $c(\{5\}) = 20$. The partitions $\{\{3, 5\}\}$ and $\{\{3\}, \{5\}\}$ give 60 and $12 + 20 = 32$, so $w(225, 23) = 32$ by Theorem A. The realising divisor is $h = f_{45} f_{75}$, with $\text{ord}(h) = \text{lcm}(45, 75) = 225$. The distinct moduli 45 and 75 have the **same** prime support $\{3, 5\}$: an admissible support can be used by more than one block. For comparison, (6.3) gives $w_0(225, 23) = \min(60, o(9) + o(25)) = 26$, since in the unconstrained problem the blocks $\{3\}$ and $\{5\}$ need no helpers.

7 Exact two-prime-power case

7.1 Statement of Theorem B

Setting and notation. Throughout §7: $N = p^a r^b$, where p and r are distinct odd primes, $a, b \geq 1$, and $\gcd(N, q) = 1$; $o_p = o(p^a)$ and $o_r = o(r^b)$; $o_{p0} = o(p)$ and $o_{r0} = o(r)$; $t_p = v_2(o_{p0})$ and $t_r = v_2(o_{r0})$, the prime-level 2-adic classes of §5.1. By (5.1) they also equal $v_2(o_p)$ and $v_2(o_r)$. $L = \text{lcm}(o_p, o_r)$.

Order of a coprime product. For coprime moduli M_1, M_2 and integer k , CRT gives $q^k \equiv 1 \pmod{M_1 M_2}$ if and only if $q^k \equiv 1$ modulo both M_1 and M_2 . Hence

$$o(M_1 M_2) = \text{lcm}(o(M_1), o(M_2)). \quad (7.1)$$

In particular $L = o(N) = \text{ord}_N(q)$, and $o(p^a r) = \text{lcm}(o_p, o_{r0})$, $o(p r^b) = \text{lcm}(o_{p0}, o_r)$.

Case	Condition	$w(N, q)$	$w_0(N, q)$
I	$t_p = t_r \geq 1$	∞	$\min(L, o_p + o_r)$
II	$t_p = t_r = 0$	$\min(L, o_p + o_r)$	$\min(L, o_p + o_r)$
III	$t_p = 0, t_r \geq 1$	$\min(L, o_p + \text{lcm}(o_{p0}, o_r))$	$\min(L, o_p + o_r)$
IV	$t_p, t_r \geq 1, t_p \neq t_r$	$\min(L, \text{lcm}(o_p, o_{r0}) + \text{lcm}(o_{p0}, o_r))$	$\min(L, o_p + o_r)$

Theorem B *Let $N = p^a r^b$ with p and r distinct odd primes, $a, b \geq 1$, and $\gcd(N, q) = 1$. Then: in every case, $w_0(N, q) = \min(L, o_p + o_r)$; $w(N, q)$ is given by the table above. The case $t_r = 0, t_p \geq 1$ is Case III with p and r interchanged, and then $w(N, q) = \min(L, o_r + \text{lcm}(o_p, o_{r0}))$.*

7.2 Proof by case analysis

Common setup. We apply Theorem A with $P = \{p, r\}$. *Partitions.* There are exactly two: the one-block partition $\{\{p, r\}\}$ and the two-block partition $\{\{p\}, \{r\}\}$. *Admissibility of supports.* By Lemma 5.1 and Corollary 5.2, a nonempty support $I \subseteq \{p, r\}$ is admissible unless all t -values on I are equal and positive. *One-block cost.* For $A = \{p, r\}$, the only helper set is $H = \emptyset$, and $m(A, \emptyset) = N$. So $c(\{p, r\}) = L$ if $\{p, r\}$ is admissible, and ∞ otherwise. *Singleton costs.* For $A = \{p\}$ there are two helper sets: $H = \emptyset$ gives $m = p^a$ and order o_p ; $H = \{r\}$ gives $m = p^a r$ and order $\text{lcm}(o_p, o_{r0})$ by (7.1).

Since o_p divides $\text{lcm}(o_p, o_{r0})$, whenever $\{p\}$ is itself admissible the empty helper set is optimal. So

$$c(\{p\}) = o_p \text{ if } \{p\} \text{ is admissible; } \text{lcm}(o_p, o_{r0}) \text{ if } \{p\} \text{ is inadmissible and } \{p, r\} \text{ is admissible; } \infty \text{ otherwise.}$$

The same holds for $c(\{r\})$ with p and r interchanged. *The minimum.* By Theorem A, $w(N, q) = \min(c(\{p, r\}), c(\{p\}) + c(\{r\}))$.

w_0 (**all cases**). By (6.3), $w_0(N, q) = \min(o(N), o(p^a) + o(r^b)) = \min(L, o_p + o_r)$. This does not depend on the t -values.

Case I ($t_p = t_r = t \geq 1$). $\{p\}$ and $\{r\}$ are inadmissible, since each carries a single positive class. $\{p, r\}$ is inadmissible, since both classes equal the same positive integer. So every divisor $m > 1$ of N is inadmissible (Corollary 5.2), and $m = 1$ is never admissible. In particular N is inadmissible. Hence $w(N, q) = \infty$ by Proposition 4.3(1). Equivalently, every block cost in Theorem A is ∞ .

Case II ($t_p = t_r = 0$). All three supports contain a prime of class 0, so all are admissible. So $c(\{p, r\}) = L$, $c(\{p\}) = o_p$ and $c(\{r\}) = o_r$. Hence $w(N, q) = \min(L, o_p + o_r)$, which is exactly the unconstrained value: no helper is ever needed.

Case III ($t_p = 0, t_r \geq 1$). $\{p\}$ and $\{p, r\}$ are admissible, because each contains p , of class 0. $\{r\}$ is inadmissible, because it has a single positive class. One block: $c(\{p, r\}) = L$. Block $\{p\}$: admissible on its own, so $c(\{p\}) = o_p$. Block $\{r\}$: $H = \emptyset$ is inadmissible. The helper set $\{p\}$ gives support $\{p, r\}$, which is admissible, with auxiliary modulus $r^b p \mid N$ and order $\text{lcm}(o_{p0}, o_r)$ by (7.1). So $c(\{r\}) = \text{lcm}(o_{p0}, o_r)$. Hence $w(N, q) = \min(L, o_p + \text{lcm}(o_{p0}, o_r))$.

Case IV ($t_p, t_r \geq 1, t_p \neq t_r$). $\{p\}$ and $\{r\}$ are inadmissible, since each has a single positive class. $\{p, r\}$ is admissible, since the two classes differ. One block: $c(\{p, r\}) = L$. Block $\{p\}$: the only admissible helper set is $\{r\}$. The auxiliary modulus is $p^a r \mid N$, with order $\text{lcm}(o_p, o_{r0})$. Block $\{r\}$: the only admissible helper set is $\{p\}$. The auxiliary modulus is $p r^b \mid N$, with order $\text{lcm}(o_{p0}, o_r)$. Hence $w(N, q) = \min(L, \text{lcm}(o_p, o_{r0}) + \text{lcm}(o_{p0}, o_r))$.

Symmetric case ($t_r = 0, t_p \geq 1$). Interchange the roles of p and r in Case III. ■

In Cases III and IV the two-block cost may be smaller or larger than L . The formulas take the minimum, and neither alternative is dominant in general.

7.3 Comparison with w_0

Theorem B makes Proposition 4.3(2), $w \geq w_0$, explicit. All three possible relations occur:

Case I: $w = \infty > w_0$. The reciprocal-free problem is infeasible. **Case II:** $w = w_0$. **Cases III and IV:** w and w_0 share the one-block term L . In the two-block term, one or both of o_p and o_r is replaced by an lcm that is divisible by it. So $w \geq w_0$, with equality exactly when the minimum in both formulas is attained at L , or when the replaced lcms coincide with o_p and o_r . The inequality can be strict (§7.4).

In particular, $w > w_0$ is not universal.

7.4 Examples

All values below come from Theorem B and agree with the verification record. (i) $(q, N) = (3, 143) = (3, 11 \cdot 13)$, Case II: $o(11) = 5$, $o(13) = 3$, $t_{11} = t_{13} = 0$, and $w = w_0 = \min(15, 5 + 3) = 8 < \text{ord}_{143}(3) = 15$. (ii) $(3, 55) = (3, 11 \cdot 5)$, Case III with the class-0 prime 11: $o(11) = 5$ ($t = 0$), $o(5) = 4$ ($t = 2$), $w = \min(20, 5 + \text{lcm}(5, 4)) = 20$ and $w_0 = \min(20, 9) = 9$, a genuine gap. (iii) $(3, 35) = (3, 5 \cdot 7)$, Case IV: $o(5) = 4$ ($t = 2$), $o(7) = 6$ ($t = 1$), $w = \min(12, \text{lcm}(4, 6) + \text{lcm}(4, 6)) = 12$ and $w_0 = \min(12, 10) = 10$. (iv) $(23, 225)$, $N = 3^2 \cdot 5^2$, Case IV (§6.6): $o(3) = 2$, $o(9) = 6$, $o(5) = 4$, $o(25) = 20$, $t_3 = 1$, $t_5 = 2$, $w = \min(60, \text{lcm}(6, 4) + \text{lcm}(2, 20)) = 32 < L = 60$, and $w_0 = \min(60, 26) = 26$.

8 Separation between w and w_0

8.1 Theorem C(i)

Theorem C (i) *Let $p \neq r$ be odd primes with $p, r \nmid q$. Suppose $t_p \neq t_r$ and $t_p, t_r \geq 1$. Then*

$$w(pr, q) = \text{lcm}(o(p), o(r)) = \text{ord}_{pr}(q), \text{ and } w_0(pr, q) \leq o(p) + o(r).$$

Proof Apply Theorem B, Case IV, with $a = b = 1$. Then $o_p = o_{p0} = o(p)$ and $o_r = o_{r0} = o(r)$, and $L = \text{lcm}(o(p), o(r)) = \text{ord}_{pr}(q)$ by (7.1). Both helper-supported singleton costs collapse: $\text{lcm}(o_p, o_{r0}) = \text{lcm}(o_{p0}, o_r) = L$. The auxiliary moduli are $p \cdot r = N$ in both cases. So the two-block partition costs $2L$. The one-block partition costs L . Hence $w(pr, q) = \min(L, 2L) = L$. The bound for w_0 is the two-singleton partition in (6.3). $\square$

8.2 An explicit family

Theorem C (ii), first part *Let p and r be primes with: $p \equiv 3 \pmod{4}$ and $p \geq 7$; $r \equiv 5 \pmod{8}$; $\gcd((p-1)/2, (r-1)/4) = 1$.*

Let q be a prime that is a primitive root modulo p and modulo r . Then

$$w(pr, q) = (p-1)(r-1)/2 \text{ and } w_0(pr, q) = p + r - 2.$$

Proof 2-adic classes. Since q is a primitive root modulo p and modulo r , neither p nor r divides q , so $\gcd(pr, q) = 1$. Also $\text{o}(p) = p - 1$ and $\text{o}(r) = r - 1$. $p \equiv 3 \pmod{4}$ gives $v_2(p-1) = 1$, and $r \equiv 5 \pmod{8}$ gives $v_2(r-1) = 2$. Hence $t_p = 1$ and $t_r = 2$: both positive and unequal. By Theorem C(i), $w(pr, q) = \text{lcm}(p-1, r-1)$.

The lcm. Write $p - 1 = 2u$ and $r - 1 = 4v$ with u, v odd, and $\gcd(u, v) = 1$ by hypothesis. Then $\gcd(p-1, r-1) = \gcd(2u, 4v) = 2 \cdot \gcd(u, 2v) = 2 \cdot \gcd(u, v) = 2$, using that u is odd. Hence $\text{lcm}(p-1, r-1) = (p-1)(r-1)/2$.

w_0 . By (6.3) with $a = b = 1$, $w_0(pr, q) = \min((p-1)(r-1)/2, (p-1) + (r-1))$. Now $(p-1)(r-1)/2 \geq (p-1) + (r-1)$ if and only if $(p-3)(r-3) \geq 4$. Since $p \geq 7$ and $r \geq 5$, $(p-3)(r-3) \geq 4 \cdot 2 = 8$. So the minimum is $p + r - 2$. □

The hypothesis $p \geq 7$ excludes $p = 3$, where $(p-3)(r-3) = 0$. For $p = 3$ one gets $w_0 = r - 1 = w$ instead.

8.3 Unboundedness over pairs (N, q)

The argument uses two classical theorems: **(PR)** For every prime ℓ , the group $(\mathbb{Z}/\ell\mathbb{Z})^\times$ is cyclic. That is, a primitive root modulo ℓ exists. **(D)** *Dirichlet's theorem on primes in arithmetic progressions*: if $\gcd(c, M) = 1$, there are infinitely many primes $\equiv c \pmod{M}$.

No result on primitive roots for a fixed base (Artin-type) is used.

Theorem C (ii), second part *There are infinitely many triples (p, r, q) satisfying the hypotheses of §8.2, and they can be chosen with $\min(p, r) \rightarrow \infty$. Consequently the set $\{w(N, q)/w_0(N, q) : N \text{ odd}, q \text{ prime}, \gcd(N, q) = 1\}$ is unbounded.*

Proof Step 1 (choice of p). By (D) with $M = 4$, there are infinitely many primes $p \equiv 3 \pmod{4}$. Choose such primes $p_1 < p_2 < \dots$ with $p_1 \geq 7$.

Step 2 (choice of r for a given p). Let $u = (p-1)/2$, which is odd. Let $\ell_1, \dots, \ell_k$ be the distinct primes dividing u (possibly $k = 0$). All ℓ_i are odd. By CRT, there is a residue class c' modulo $M' = 8\ell_1 \cdots \ell_k$ with $c' \equiv 5 \pmod{8}$ and $c' \equiv 2 \pmod{\ell_i}$ for each i . $\gcd(c', M') = 1$, because 5 is odd and $2 \not\equiv 0 \pmod{\ell_i}$. By (D), there are infinitely many primes $r \equiv c' \pmod{M'}$. Choose one with $r > p$. Then $r \equiv 5 \pmod{8}$. Also $r - 1 \equiv 1 \pmod{\ell_i}$, so no ℓ_i divides $r - 1$, and hence none divides $(r-1)/4$. Therefore $\gcd(u, (r-1)/4) = 1$.

Step 3 (choice of q for a given pair (p, r)). By (PR), choose primitive roots g_p modulo p and g_r modulo r . By CRT, there is c modulo pr with $c \equiv g_p \pmod{p}$ and $c \equiv g_r \pmod{r}$. Then $\gcd(c, pr) = 1$. By (D), there are infinitely many primes $q \equiv c \pmod{pr}$. For any such q : $q \equiv g_p \pmod{p}$, so $\text{o}(p) = p - 1$, and similarly $\text{o}(r) = r - 1$. So q is a primitive root modulo both p and r . The multiplicative order modulo a prime depends only on the residue class.

Step 4 (the ratio). For each k , Steps 2 and 3 give a triple (p_k, r_k, q_k) satisfying the hypotheses of §8.2, with $r_k > p_k$. By §8.2, with $s = \min(p_k, r_k) = p_k$ and $M = \max(p_k, r_k)$:

$$w/w_0 = (p_k - 1)(r_k - 1) / (2(p_k + r_k - 2)) \geq (s-1)(M-1) / (2 \cdot 2(M-1)) = (s - 1)/4,$$

using $p_k + r_k - 2 \leq 2(M - 1)$. Since $s = p_k \rightarrow \infty$, the ratio is unbounded along $(N_k, q_k) = (p_k, r_k, q_k)$. □

Fixed- q scope remark. Theorem C(ii) establishes unbounded separation of w from w_0 over pairs (N, q) , with the prime q chosen after, and depending on, $N = \text{pr}$. It does not establish unboundedness of $w(N, q)/w_0(N, q)$ for any fixed q .

9 Application to quantum synchronizable codes

9.1 Saturated cyclic QSC framework

Throughout §9–§11, $\mathbf{gcd}(N, q) = 1$, and $C \subsetneq D$ is a pair as in §2.4: $g_D \mid g_C$, $g_D \neq g_C$, and $C^\perp \subseteq C$. The synchronisation factor is $h = g_C/g_D$, with $s = \deg h$ and $B = \deg g_D$, so that $\deg g_C = B + s$, $k_C = N - \deg g_C$, $k_D = \dim D = N - B$, and $K = 2k_C - N = N - 2 \deg g_C$ by (2.3). As before, $d(D)$ is the **classical** minimum distance of D , not a quantum distance. By (2.2) the tolerance satisfies $a_l + a_r < \text{ord}(h) \leq N$, and the pair is *saturated* when $\text{ord}(h) = N$.

9.2 w as a saturation-cost lower bound

Proposition 9.1 (saturation cost) *Assume $\mathbf{gcd}(N, q) = 1$ and the setting of §9.1. If $\text{ord}(h) = N$, then h is feasible for $w(N, q)$ in Definition 4.1. In particular,*

$$s = \deg h \geq w(N, q).$$

Proof h divides g_C , which divides $x^N - 1$. By Lemma 2.3, $g_C g_C^*$ divides $x^N - 1$. Since $\mathbf{gcd}(N, q) = 1$, $x^N - 1$ is squarefree, so $\mathbf{gcd}(g_C, g_C^*) = 1$. As $h \mid g_C$ and $h^* \mid g_C$, this gives $\mathbf{gcd}(h, h) = 1$. Together with $\text{ord}(h) = N$, h is feasible for $w(N, q)$, and the inequality follows from Definition 4.1. $\square$

Consequences. By Proposition 4.3(1), saturation in this framework requires N to be admissible, and $w(N, q)$, which depends only on (N, q) , is the least possible degree of a saturating factor.

9.3 Consequence for the parameter inequality

Corollary 9.2 *Under the hypotheses of Proposition 9.1,*

$$K + 2d(D) \leq N + 2 - 2s \leq N + 2 - 2w(N, q). \quad (9.1)$$

Proof $D \neq \{0\}$, because $C^\perp \subseteq C \subsetneq D$ forces $C \neq \{0\}$. So Lemma 2.4 applied to D gives $d(D) \leq N - k_D + 1 = B + 1$. $K = N - 2 \deg g_C = N - 2B - 2s$. Hence $K + 2d(D) \leq N - 2B - 2s + 2B + 2 = N + 2 - 2s$. The second inequality in (9.1) is Proposition 9.1. $\square$

The middle term $N + 2 - 2s$ depends on the construction, through s . The right-hand term depends only on (N, q) .

9.4 Scope of the QSC interpretation

Within the cyclic framework, w lower-bounds the degree of a saturating factor (Proposition 9.1) and enters the elementary inequality (9.1), which follows from the Singleton bound applied to D . We do **not** claim that w determines K , $d(D)$, the quantum error-correcting capability, or the set of achievable QSC parameters; that (9.1) is sharp or improves Singleton-type bounds in general; or that these statements extend beyond $\gcd(N, q) = 1$. In particular, repeated-root constructions such as those of [5] are outside the scope of Proposition 9.1 and are not treated here.

10 Deficit decomposition

10.1 Definition of the deficit Δ

Assume the hypotheses of Proposition 9.1 ($\gcd(N, q) = 1$, the setting of §9.1, and $\text{ord}(h) = N$). The *deficit* of the saturated pair (C, D) is

$$\Delta = [N + 2 - 2w(N, q)] - [K + 2d(D)].$$

So Δ is the slack in the right-hand inequality of (9.1), and $\Delta \geq 0$ by Corollary 9.2.

10.2 Exact deficit decomposition

Proposition 10.1 (deficit decomposition) *Under the hypotheses of Proposition 9.1,*

$$\Delta = 2(s - w(N, q)) + 2(B + 1 - d(D)), \quad (10.1)$$

and both summands are non-negative.

Proof Substituting $K = N - 2B - 2s$ into the definition of Δ gives

$$\Delta = N + 2 - 2w - N + 2B + 2s - 2d(D) = 2(s - w) + 2(B + 1 - d(D)).$$

$s - w \geq 0$ by Proposition 9.1. $B + 1 - d(D) \geq 0$ by Lemma 2.4 applied to D , as in the proof of Corollary 9.2. □

The hypothesis $\gcd(N, q) = 1$ enters only through Proposition 9.1.

10.3 Interpretation of the two slack terms

The decomposition (10.1) separates two sources of slack. The **structural excess** $2(s - w)$ measures how far the degree of the actual saturating factor h exceeds the intrinsic minimum $w(N, q)$. The **coding-side slack** $2(B + 1 - d(D))$ is twice the Singleton defect of the classical code D . The identity (10.1) implies Corollary 9.2 but is not a new coding bound. It does not assert that either slack term can be reduced, or reduced independently of the other: lowering s changes D , and hence $d(D)$. A positive structural excess is not evidence that a construction is inferior; K , $d(D)$ and

the tolerance must be assessed jointly, together with the quantum distances, which (10.1) does not involve.

11 Calibration with the published construction

11.1 Calibration scope

We calibrate the constructions of Li and Zhu [4] against w . In [4] the code length is $N = 2n$ with $\gcd(n, q) = 1$ and q odd, so $\gcd(N, q) = 1$ and Propositions 9.1 and 10.1 apply.

Structure of the codes in [4]. [4] builds composite codes of length $2n$ from a cyclic code of length n (generator dividing $x^n - 1$) and a negacyclic code of length n (generator dividing $x^n + 1$), via the $(u+v \mid u-v)$ construction. For odd q , $c(x) \mapsto (c \bmod (x^n - 1), c \bmod (x^n + 1))$ is a ring isomorphism $F_q[x]/(x^{2n} - 1) \rightarrow F_q[x]/(x^n - 1) \oplus F_q[x]/(x^n + 1)$. Under this isomorphism, and up to the scalar $1/2$, the $(u+v \mid u-v)$ composite of $\langle g_1 \rangle$ and $\langle g_2 \rangle$ is the cyclic code of length $2n$ with generator $g_1 g_2$. Accordingly, in [4]’s notation, $g_C = g_1 g_2$ and $g_D = g_3 g_4$, and

$$h = (g_1/g_3)(g_2/g_4),$$

where $g_3 \mid g_1$ divides $x^n - 1$ and $g_4 \mid g_2$ divides $x^n + 1$.

What calibration means here. Calibration compares the construction cost s with the intrinsic cost w and reports the decomposition (10.1); it does not rank [4] against other constructions. The constructions of [4] are used exactly as stated there, and printed inconsistencies are recorded explicitly rather than corrected silently.

11.2 Class 1: $N = 2n$, n an odd prime

Theorem 11.1 (calibration of [4], Class 1) *Hypotheses: q is an odd prime power, and n is an odd prime with $n \nmid q$; $-1 \notin \langle q \rangle \bmod 2n$; $o = \text{ord}_n(q)$; (C, D) is a Class-1 pair of [4], with [4]’s strict-nesting hypotheses on both sides, so that g_1/g_3 and g_2/g_4 are nonconstant; $\text{ord}(h) = 2n$.*

Then: (i) $\text{ord}_{2n}(q) = o$, and $w(2n, q) = w_0(2n, q) = o$. (ii) $s \geq 2w(2n, q)$. Equality $s = 2w$ holds if and only if h contains exactly one irreducible factor of Φ_n and exactly one irreducible factor of Φ_{2n} . (iii) $\Delta \geq 2w(2n, q)$. Equality $\Delta = 2w$ holds if and only if $s = 2w$ and $d(D) = B + 1$.

Proof (i) Orders and the value of w . Since q is odd, $q^j \equiv 1 \pmod{2}$ for all j . By CRT, $\text{ord}_{2n}(q) = \text{ord}_n(q) = o$. For the same reason, $-1 \in \langle q \rangle \bmod 2n$ if and only if $-1 \in \langle q \rangle \bmod n$. So both n and $2n$ are admissible. The divisors of $2n$ that are ≥ 3 are exactly n and $2n$. By Proposition 4.2, an admissible cover of $2n$ must contain $2n$, because $\text{lcm}\{n\} = n$. So $w(2n, q) = o(2n) = o$. For w_0 , the covers are $\{2n\}$ (cost o), $\{n, 2\}$ (cost $o + 1$), and supersets of these. So $w_0(2n, q) = o$.

(ii) The degree of h . Put $h_1 = g_1/g_3$ and $h_2 = g_2/g_4$. Both are nonconstant by hypothesis. By Lemma 2.3 and squarefreeness, $g_C = g_1 g_2$ is reciprocal-free. So it has no self-reciprocal irreducible factor, and in particular $x - 1 \nmid g_1$ and $x + 1 \nmid g_2$. Since $x^n - 1 = (x - 1)\Phi_n$

and $x^n + 1 = (x + 1)\Phi_{2n}$, it follows that $h_1 \mid \Phi_n$ and $h_2 \mid \Phi_{2n}$. Every irreducible factor of Φ_n has order n and degree $o(n) = o$. Every irreducible factor of Φ_{2n} has order $2n$ and degree $o(2n) = o$ (§2.2). Hence $\deg h_1$ and $\deg h_2$ are positive multiples of o , and $s = \deg h_1 + \deg h_2 \geq 2o = 2w$. Equality holds if and only if $\deg h_1 = \deg h_2 = o$, that is, if and only if each of h_1 and h_2 is a single irreducible factor.

(iii) *The deficit.* By (10.1) and (ii),

$$\Delta = 2(s - w) + 2(B + 1 - d(D)) \geq 2(2w - w) = 2w.$$

Since both summands of (10.1) are non-negative, equality requires $s = 2w$ and $d(D) = B + 1$. □

Remarks on Theorem 11.1. Index-level equality needs an extra assumption. [4] writes g_1 and g_3 as products of half-factors $f_{n,j}$, and g_2 and g_4 as products of half-factors y_j . The y_j are irreducible, by [4]’s count of the factors of Φ_{2n} . The equality condition in (ii) can be restated as “exactly one index with a gap on each side” **only under the additional assumption** that the $f_{n,j}$ are irreducible. That assumption is not established here. **Source of the factor 2.** The bound $s \geq 2w$ reflects the construction: [4] requires a nontrivial factor on both the cyclic side (h_1) and the negacyclic side (h_2). Only h_2 contributes order $2n$. The factor h_1 has order n , contributes to the degree, and is not needed for $\text{ord}(h) = 2n$. A single irreducible factor of Φ_{2n} is reciprocal-free, has order $2n$, and has degree w . Theorem 11.1 does not assert that w itself doubles, nor that a construction with $s = w$ is available within [4]’s hypotheses. **Parity of N .** Theorem 11.1 is proved directly for even $N = 2n$. It does not rely on Theorem A, which concerns odd N .

11.3 Numerical calibration of the three examples

The following values are computed: $d(D)$ (exact) is the classical minimum distance of the cyclic code D with generator $g_D = g_3 g_4$. It was computed exactly by parity-check rank. The distances printed in [4] are lower bounds from the component distance formula. For Example 3, [4] prints g_1 and g_3 containing the factor $m_0 = x - 1$. That factor is self-reciprocal, which is incompatible with dual containment, and it contradicts [4]’s printed dimensions [40,32] and [40,36]. The values below use the factor lists without m_0 ; these reproduce all printed dimensions and the printed $h = m_4 m_5$. Examples 1 and 2 belong to Class 1. Example 3 belongs to [4]’s BCH class (Class 2), so for it the values come from direct computation, not from Theorem 11.1.

In Table 1, $w_0 = w = \text{ord}_N(q)$, $N + 2 - 2w = 22, 74, 74$ and $K + 2d(D) = 10, 54, 56$, $B = \deg g_D$, $s = \deg h$, “ $d(D)$ [4]” is the lower bound printed in [4], and “ $d(D)$ ” is the exact value.

What the table shows. In all three examples $s = 2w$, the equality case of Theorem 11.1(ii) for Examples 1 and 2. The deficit splits roughly equally between structural excess and coding-side slack. Examples 1 and 2 satisfy $\Delta \geq 2w$, as Theorem 11.1(iii) requires. The inequality is strict because $d(D) < B + 1$.

Table 2 Calibration of the three examples of [A] (computed values; in all three cases $\deg g_C = B + s$ and $\text{ord}(h) = N$)

	Class	(q, N)	$w_0 = w$	B	s	K	$d(D)$ [A]	$d(D)$	Δ	$2(s - w)$	$2(B + 1 - d(D))$
Ex. 1	1	$(3, 26)$	3	6	6	2	3	4	12	6	6
Ex. 2	1	$(37, 82)$	5	10	10	42	4	6	20	10	10
Ex. 3	2	$(3, 80)$	4	8	8	48	3	4	18	8	10

11.4 Class 2 computational observations

For the BCH class of [4] (Class 2; $n = (q^{2^m} - 1)/(q - 1)$, $N = 2n$), whose general index rule is only partially legible in the available source, we used a reconstructed rule (described in Online Resource 1). For $(q, m) = (3, 2), (5, 2), (7, 2), (3, 3)$, with $w_0 = w = \text{ord}_N(q) = 4, 4, 4, 6$, 31,716 of the 31,742 parameter tuples examined are saturated, and every saturated tuple observed has $s > w$. These observations are not a theorem and not exhaustive; tuples with $\text{ord}(h) < N$ may be excluded by the exact hypotheses of [4]. The observed minimum $s = 6$ for $(q, m) = (5, 2)$ is below $2w = 8$, so $s \geq 2w$ is not claimed for Class 2.

Calibration conclusions. For every calibrated instance of [4], $w(N, q) = w_0(N, q) = \text{ord}_N(q)$, and the saturating factor of [4] is feasible for w (Proposition 9.1). In Class 1 the construction cost satisfies $s \geq 2w$, and hence $\Delta \geq 2w$ (Theorem 11.1); the excess comes from the requirement in [4] of nontrivial factors on both the cyclic and negacyclic sides. In the three worked examples $s = 2w$, $\Delta = 12, 20, 18$ splits as shown in §11.3, and the exact classical distances $d(D)$ exceed the lower bounds printed in [4]. Class 2 is calibrated only empirically, under a reconstructed index rule. None of these statements implies that the codes of [4] are suboptimal: K , $d(D)$, the tolerance and the quantum capabilities must be assessed jointly. The w -calibration does not apply to the repeated-root constructions of [5].

12 Scope, limitations and computational verification

12.1 Scope of the structural results

The exact ranges of the structural results are those stated in their hypotheses and summarised in §1.6: Lemma 5.1 and Corollary 5.2 for odd $m \geq 3$ with $\gcd(m, q) = 1$ (no parity assumption on q); Theorem A for odd N ; Theorem B for $N = p^a r^b$ with distinct odd primes; Theorem C for $N = \text{pr}$, part (ii) for an explicit family in which the prime q depends on (p, r) ; Propositions 9.1 and 10.1 and Corollary 9.2 for saturated cyclic QSC pairs with $\gcd(N, q) = 1$, for N of either parity; and Theorem 11.1 for $N = 2n$ with q odd and n an odd prime. The paper does **not** claim a support-reduction theory for general even N , a closed formula for three or more distinct prime factors beyond the finite minimisation of Theorem A, unboundedness of w/w_0 for any fixed q , or any asymptotic, density or distribution statement about w/w_0 .

12.2 Semisimple restriction and repeated-root exclusion

Every statement involving $w(N, q)$ assumes $\gcd(N, q) = 1$. The hypothesis enters twice: $x^N - 1$ is then squarefree, which gives the factorisation and degree formulas of §2.2 and the lcm-cover form (Proposition 4.2), and dual containment then implies that the synchronisation factor is reciprocal-free (Proposition 9.1). When $\gcd(N, q) > 1$, dual containment still means $g_C g_C^* \mid x^N - 1$ (Lemma 2.3), but $x^N - 1$ has repeated factors, so a divisor of g_C may contain self-reciprocal factors or reciprocal pairs with bounded multiplicity; $\gcd(h, h^*) = 1$ is then **not** the correct constraint on saturating factors, and $w(N, q)$ is not the relevant saturation cost. For this reason the constructions of [5] are not calibrated by w : their lengths are divisible by the characteristic (for example $N = 12p^s$ over a field of characteristic p), and their synchronisation factors contain powers of self-reciprocal factors and reciprocal pairs. A saturation-cost invariant for repeated-root lengths would require a different definition; none is proposed here, and no QSC parameter frontier is derived.

12.3 Computational verification

All numbered results are proved; the computations only check implementations and stress-test the statements on finite ranges, and zero failures is not offered as evidence beyond the tested inputs. The values w and w_0 were computed by a dynamic programme over the lcm-cover form (Proposition 4.2), cross-checked against an independent brute force over sets of q -cyclotomic cosets. A first implementation of the enumerator for Theorem A reported 7 spurious failures (e.g. $(q, N) = (23, 225)$) because it used each prime support at most once, whereas Theorem A allows distinct blocks to share a support (§6.6). After correction, 409,075 checks over prime powers $q < 128$ were run with zero failures, covering Lemma 5.1, Theorems A, B, C and formula (6.3), and Theorem 11.1(i), including even q , repeated prime powers, three and four distinct primes, mixed 2-adic classes and shared supports. The ranges and counts are listed in Online Resource 1; the code is Online Resource 2. For the instances from [4], generator degrees, $\text{ord}(h)$, dual containment and exact $d(D)$ were recomputed directly (§11.3).

12.4 Literature and novelty scope

The unconstrained quantity $w_0(N, q)$ is classical [1, 3] and belongs to the circle of prescribed-order and prescribed-period problems over finite fields. In the literature search performed for this work, no invariant equivalent to $w(N, q)$ was identified; the search did not cover every database (MathSciNet was not accessible, and zbMATH Open was used instead), so no absolute priority claim is made, either for the invariant or for Theorems A, B and C. Lemma 5.1 is known in substance, and formula (6.3) is a direct consequence of classical facts.

Limitations and open directions. Outside the scope of this paper are: a structural theory of $w(N, q)$ for even N , where $(\mathbb{Z}/2^a\mathbb{Z})^\times$ is not cyclic for $a \geq 3$; closed formulas for three or more distinct prime factors; a sharper description of when $w = w_0$; the behaviour of w/w_0 for fixed q ; the interaction of w with explicit optimisation of QSC parameters; and a saturation-cost invariant for repeated-root lengths.

13 Conclusion

For $\gcd(N, q) = 1$ we studied the reciprocal-free restriction $w(N, q)$ of the classical minimal degree $w_0(N, q)$. Admissibility for odd moduli is governed by the 2-adic classes t_p (Lemma 5.1); for odd N , Theorem A locates the difference between w and w_0 precisely in the helper costs; Theorem B makes this explicit for $N = p^a r^b$; and Theorem C gives arbitrarily large separation over pairs (N, q) with q varying. In the semisimple cyclic QSC framework, w bounds the degree of every saturating factor, giving $K + 2d(D) \leq N + 2 - 2w$ and the decomposition (10.1); in Class 1 of [4] the construction forces $s \geq 2w$, a property of that construction rather than a statement about code quality. No fixed- q , asymptotic or frontier statement is made, the support-reduction theory is established for odd N only, and repeated-root constructions require a different notion of saturation cost.

Supplementary information. Online Resource 1 (ESM_1.pdf): verification record and supplementary proofs (Sects. 2.3, 11.4, 12.3). Online Resource 2 (ESM_2.zip): Python code.

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Data availability. No datasets were generated or analysed; all values are reproducible with Online Resource 2. [AUTHOR TO CONFIRM]

Code availability. The Python code used for the computations is provided as Online Resource 2 [and at: INSERT REPOSITORY LINK IF DEPOSITED].

Use of AI tools. [AUTHOR TO COMPLETE: describe truthfully any use of large language models, as required by Springer Nature policy.]

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Author contributions. [INSERT]

References

- [1] Lidl, R., Niederreiter, H.: Finite Fields, 2nd edn. Cambridge University Press, Cambridge (1996). <https://doi.org/10.1017/CBO9780511525926>
- [2] Fujiwara, Y., Tonchev, V.D., Wong, T.W.H.: Algebraic techniques in designing quantum synchronizable codes. Phys. Rev. A **88**(1), 012318 (2013) <https://doi.org/10.1103/PhysRevA.88.012318>

- [3] Bush, M.R., Quijada, D.: Period sets of linear recurrences over finite fields and related commutative rings. *Involve* **14**(3), 361–376 (2021) <https://doi.org/10.2140/involve.2021.14.361>
- [4] Li, Z., Zhu, S.: Two classes of quantum synchronizable codes. *Int. J. Theor. Phys.* **61**(10), 251 (2022) <https://doi.org/10.1007/s10773-022-05163-1>
- [5] Du, C., Ma, Z., Luo, L., Huang, D., Wang, H.: On a family of quantum synchronizable codes based on the $(\lambda(u+v)|u-v)$ construction. *IEEE Access* **8**, 8449–8458 (2020) <https://doi.org/10.1109/ACCESS.2019.2963289>
- [6] Fujiwara, Y.: Block synchronization for quantum information. *Phys. Rev. A* **87**(2), 022344 (2013) <https://doi.org/10.1103/PhysRevA.87.022344>
- [7] Fujiwara, Y., Vandendriessche, P.: Quantum synchronizable codes from finite geometries. *IEEE Trans. Inf. Theory* **60**(11), 7345–7354 (2014) <https://doi.org/10.1109/TIT.2014.2357029>
- [8] Xie, Y., Yuan, J., Fujiwara, Y.: Quantum synchronizable codes from quadratic residue codes and their supercodes. In: *Proc. 2014 IEEE Inf. Theory Workshop (ITW 2014)*, pp. 172–176 (2014). <https://doi.org/10.1109/ITW.2014.6970815>
- [9] Xie, Y., Yang, L., Yuan, J.: q-ary chain-containing quantum synchronizable codes. *IEEE Commun. Lett.* **20**(3), 414–417 (2016) <https://doi.org/10.1109/LCOMM.2015.2512261>
- [10] Guenda, K., La Guardia, G.G., Gulliver, T.A.: Algebraic quantum synchronizable codes. *J. Appl. Math. Comput.* **55**(1–2), 393–407 (2017) <https://doi.org/10.1007/s12190-016-1042-7>
- [11] Luo, L., Ma, Z., Lin, D.: Two new families of quantum synchronizable codes. *Quantum Inf. Process.* **18**(9), 277 (2019) <https://doi.org/10.1007/s11128-019-2389-0>
- [12] Luo, L., Ma, Z.: Non-binary quantum synchronizable codes from repeated-root cyclic codes. *IEEE Trans. Inf. Theory* **64**(3), 1461–1470 (2018) <https://doi.org/10.1109/TIT.2018.2795479>
- [13] Du, C., Ma, Z., Liu, Y.: Quantum synchronizable codes from repeated-root quasi-cyclic codes. *Comput. Appl. Math.* **42**(4), 161 (2023) <https://doi.org/10.1007/s40314-023-02298-7>
- [14] Calderbank, A.R., Rains, E.M., Shor, P.W., Sloane, N.J.A.: Quantum error correction via codes over $GF(4)$. *IEEE Trans. Inf. Theory* **44**(4), 1369–1387 (1998) <https://doi.org/10.1109/18.681315>
- [15] Ketkar, A., Klappenecker, A., Kumar, S., Sarvepalli, P.K.: Nonbinary stabilizer

- codes over finite fields. *IEEE Trans. Inf. Theory* **52**(11), 4892–4914 (2006) <https://doi.org/10.1109/TIT.2006.883612>
- [16] Aly, S.A., Klappenecker, A., Sarvepalli, P.K.: On quantum and classical BCH codes. *IEEE Trans. Inf. Theory* **53**(3), 1183–1188 (2007) <https://doi.org/10.1109/TIT.2006.890730>
 - [17] Blackford, T.: Negacyclic duadic codes. *Finite Fields Appl.* **14**(4), 930–943 (2008) <https://doi.org/10.1016/j.ffa.2008.05.004>
 - [18] Kai, X., Zhu, S.: New quantum MDS codes from negacyclic codes. *IEEE Trans. Inf. Theory* **59**(2), 1193–1197 (2013) <https://doi.org/10.1109/TIT.2012.2220519>
 - [19] Massey, J.L.: Linear codes with complementary duals. *Discrete Math.* **106–107**, 337–342 (1992) [https://doi.org/10.1016/0012-365X\(92\)90563-U](https://doi.org/10.1016/0012-365X(92)90563-U)
 - [20] Yang, X., Massey, J.L.: The condition for a cyclic code to have a complementary dual. *Discrete Math.* **126**(1–3), 391–393 (1994) [https://doi.org/10.1016/0012-365X\(94\)90283-6](https://doi.org/10.1016/0012-365X(94)90283-6)
 - [21] Sloane, N., Thompson, J.: Cyclic self-dual codes. *IEEE Trans. Inf. Theory* **29**(3), 364–366 (1983) <https://doi.org/10.1109/TIT.1983.1056682>
 - [22] Jia, Y., Ling, S., Xing, C.: On self-dual cyclic codes over finite fields. *IEEE Trans. Inf. Theory* **57**(4), 2243–2251 (2011) <https://doi.org/10.1109/TIT.2010.2092415>
 - [23] Moree, P.: On the divisors of $a^k + b^k$. *Acta Arith.* **80**(3), 197–212 (1997) <https://doi.org/10.4064/aa-80-3-197-212>
 - [24] Darafsheh, M.R.: Order of elements in the groups related to the general linear group. *Finite Fields Appl.* **11**(4), 738–747 (2005) <https://doi.org/10.1016/j.ffa.2004.12.003>
 - [25] Schmutz, E.: The order of a typical matrix with entries in a finite field. *Israel J. Math.* **91**(1–3), 349–371 (1995) <https://doi.org/10.1007/BF02761656>